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Knut Erik Teigen Giljarhus

knut.e.giljarhus@uis.no

University of Stavanger

Luca Oggiano

NablaFlow AS

Method Article

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Machine learning surrogate model for athlete aerodynamics with application to cycling

Knut Erik Teigen Giljarhus[1,2*] and Luca Oggiano[2]

[1*]Department of Mechanical and Structural Engineering and Materials Science, University of Stavanger, Stavanger, Norway.
[2]NablaFlow AS, Stavanger, Norway.

*Corresponding author(s). E-mail(s): knut.e.giljarhus@uis.no;

Abstract

Aerodynamic drag dominates the resistive forces in many sports at racing speeds, and small athlete posture changes can produce practically meaningful changes in performance. Standard approaches like field testing, wind-tunnel testing and computational fluid dynamics can be reliable, but time-consuming. Recent advances in scientific machine learning have enabled surrogate models that predict flow quantities at a fraction of the computational cost of a full computational fluid dynamics simulation. However, most applications currently focus on industrial geometries, such as cars or airplanes, where the surface is morphed to generate large datasets for training. It remains unclear how well these methods transfer to athlete geometries where variability is dominated by articulated pose changes rather than smooth shape morphing. In this work, a dataset for cyclist aerodynamics is generated by combining 12 scanned athlete geometries with 20 postures per athlete. Simulations are performed for these 240 geometries, which are then used to train a state-of-the-art surrogate model. An investigation is made on the generalization of the model to an unseen geometry and the balance between number of positions and number of unique geometries in the dataset. The surrogate reproduces the drag area with good agreement, with all predictions within $\pm 10\,\%$ and a mean absolute percentage error of $2.69\,\%$. The results also indicate that increasing the number of unique athlete geometries improves performance more effectively than increasing the number of postures per athlete at fixed dataset size. These results support the feasibility of scalable surrogate models for flow

over athletes, enabling posture screening and comparative assessment at interactive speeds. This may open up new workflows and new application areas of drag calculations in the domain of sports aerodynamics.

Keywords: Computational fluid dynamics, machine learning, surrogate model, athlete posture, aerodynamics

1 Introduction

Aerodynamic drag is one of the key determinants of performance in high-speed sports, such as cycling and downhill skiing. Athlete posture has a direct effect on drag and, consequently, on power demand at race speeds. This makes athlete aerodynamics a central topic in sports engineering research.

Field tests, wind-tunnel testing and computational fluid dynamics (CFD) have been the dominant tools for investigating and improving athlete aerodynamics [1, 2]. Field testing captures the athlete in a fully realistic operating environment, but knowing the exact boundary conditions is difficult and environmental variability is hard to control. Wind tunnels can provide a controlled and realistic environment, but are expensive and time-constrained. CFD can provide richer field information, but high-fidelity CFD remains computationally demanding.

An emerging alternative is machine learning (ML) surrogate modeling, where a model is trained on a dataset of simulations to approximate the CFD mapping from geometry and boundary conditions to aerodynamic quantities of interest. Over the last years, deep learning surrogates have become a computationally efficient alternative in science and engineering [3–5]. CFD surrogate modeling, however, offer additional challenges: industrial and high-fidelity problems can involve large meshes, the number of simulation samples is typically limited, and the physics is strongly geometry-dependent [6].

One of the earlier demonstrations that ML could support aerodynamic workflows was using a custom shape and field parameterization coupled with Gaussian-process regression[7]. This later led to several different approaches for aerodynamics-oriented surrogate approaches. Several authors have used grid-based encodings of geometry, for example signed distance fields (SDFs), together with convolutional architectures such as 3D U-Nets [8, 9]. Another class of methods learns directly on meshes or point clouds using graph or message-passing formulations [10, 11]. In parallel, the scientific ML community developed operator learning methods that aim to learn mesh-free mappings between function spaces. Examples include DeepONets [12], neural operators and Fourier neural operators (FNO) [13, 14], and geometry-aware extensions that better accommodate complex domains [15]. These approaches have accelerated progress in large-scale simulation surrogates in general, including applications relevant to external aerodynamics [16]. However, a challenge for both operator-learning and mesh-based methods is scalability to large, high-fidelity geometries and fields. Many approaches require substantial downsampling of the underlying simulation mesh, which can degrade accuracy and generalization, and can be particularly problematic when the flow features of interest are localized or geometry-induced [6, 17, 18]. A further practical issue is that many models implicitly depend on the spatial structure of the training discretization and can lose accuracy when evaluated on different point samplings, for example uniform point clouds or alternative meshes [17].

Recent architectures aim to improve on these limitations by separating geometry encoding from field prediction, and by enabling decoding at arbitrary query points. One example is DoMINO (Decomposable Multi-scale Iterative

Neural Operator), a point-cloud based surrogate that learns multi-scale geometry representations, extracts local encodings, and predicts surface and volume fields via dynamically constructed local computational stencils [17].

Although these technologies are promising, it is important to evaluate their performance across different fields and disciplines. In this context, athlete aerodynamics represents a distinct challenge. Rather than morphing a single smooth surface family, such as in the DrivAer dataset [19, 20], athlete geometries feature large pose variations with articulated limbs, self-occlusion, and posture-dependent flow separation and wake dynamics.

In this work, a state-of-the-art surrogate model is trained on a large CFD dataset for flow over cyclists, generated using the virtual skeleton method [21] to efficiently produce posture variations. The motivation is not to replace wind tunnel experiments or CFD simulations, but to enable new workflows and additional applications by reducing inference time by orders of magnitude. Concretely, the following questions are proposed: accepting the inherent uncertainties and limitations of the underlying CFD model, can an ML surrogate reproduce the CFD outputs with significantly reduced runtime, and does it generalize to (i) new postures within the training distribution and (ii) athletes not seen during training. The trade-off between the number of athlete geometries and the number of postures per athlete is investigated, with the objective of establishing practical guidance for dataset design in sports-aerodynamics surrogate modeling.

The novel aspects of the presented methodology can be summarized as:

- Use of a virtual skeleton method to generate a large training dataset for flow over cyclists with systematic posture variability.
- Demonstration of the viability of modern scalable neural surrogate methods for sports aerodynamics

- Investigation of generalization to both unseen postures and unseen athletes outside the training dataset.
- Exploration of the balance between the number of athlete geometries and the number of postures per athlete when constructing training datasets for sports aerodynamics.

2 Methods

2.1 Geometries and dataset

The dataset used consisted of 12 unique, 3D-scanned athlete geometries, as shown in Fig. 1-top.

It is a challenge to define a suitable collection of athletes of sufficient generality. This will depend on the ultimate purpose of the ML surrogate model. In this work, the study population consisted of 5 female (height 169 ± 7 cm [164–180], BMI of 19.9 ± 0.5 kg m^{-2} [19.5–20.8]) and 7 male participants (height 178 ± 6 cm [170–185], BMI of 21.0 ± 0.8 kg m^{-2} [20.0–21.8]).

The geometries included both vertical and horizontal crank position, and alternate legs in the upper and lower position. With respect to equipment, all athletes were digitally equipped with the Bambino helmet from KASK (Chiuduno, Italy).

From each of these geometries, 20 positions were generated using the virtual skeleton method. An example of the 20 positions for a single athlete is shown in Fig. 1-bottom.

This gives a total of 240 geometries. The geometries were split into 70 % training, 20 % validation and 10 % testing. Several versions of the split were tested without observing significant differences in the results. Additionally, the Generic Cyclist Model [22–24] was also used to evaluate whether the surrogate

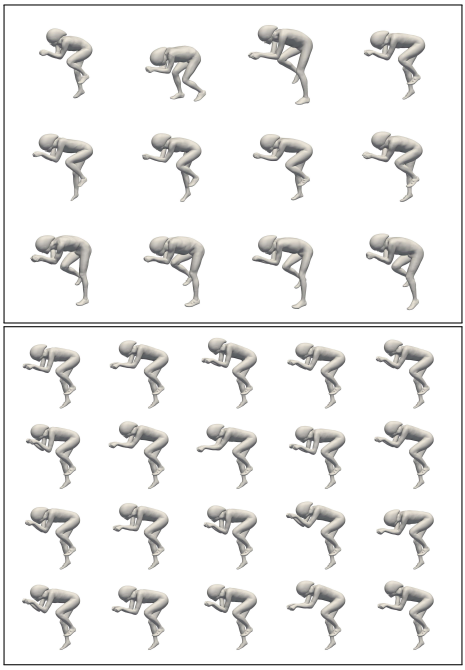


Fig. 1 Top: Side view of all 12 models used for the training. Bottom: All 20 positions created from a single geometry using the virtual skeleton methodology.

model can generalize to geometries outside the training dataset. From this model, 20 additional positions were generated and added to the test set.

2.2 Computational fluid dynamics simulations

The CFD setup mirrored the configuration previously validated for athlete–posture aerodynamics [25]. The simulations were run using the online simulation platform AeroCloud (NablaFlow, Stavanger, Norway). This allowed efficient generation and simulation of the large datasets using Amazon AWS cloud infrastructure. The simulation software was OpenFOAM-v2212. Turbulence was modeled with the k-omega SST closure, pressure–velocity coupling used the SIMPLE algorithm, and second-order schemes were applied for spatial discretization of all transported variables.

The computational domain was sized to maintain a small blockage ratio ($\lesssim 3\%$), with the inlet placed 5 body lengths upstream and the outlet 10 body lengths downstream. The model was centered laterally and vertically.

A uniform inflow speed of 15 m/s, representative of time–trial conditions was prescribed, together with a free-stream turbulence intensity of 0.5 %. A fixed static pressure was set at the outlet with zero–gradient conditions for the remaining variables. Lateral and top boundaries employed slip (symmetry) conditions. No–slip walls were imposed on the rider surfaces with smooth–wall roughness parameters.

Unstructured meshes were generated using snappyHexMesh with localized refinement around the rider and in the near wake. Anisotropic prism layers (typically 14–16 layers, growth factor ≤ 1.2) were extruded from the surface to resolve the boundary layer. The first–cell height was chosen to obtain $y^+ \lesssim 2$ over the entire body, with a first cell-height of 2 µm. The cell lengths outside the prism layer near the rider were 4 mm, and the overall cell counts were around 10 million. Each case was advanced until residuals plateaued and integral forces stabilized, and the running mean of the drag force reached a convergence level of 1×10^{-4}.

2.3 Machine learning surrogate model

The chosen ML surrogate model was the DoMINO architecture [17]. DoMINO was developed for large-scale external aerodynamics problems and released as open-source by NVIDIA (Santa Clara, United States) through the PhysicsNeMo framework [26].

In this model, the geometry is represented as a surface point cloud augmented with auxiliary geometric descriptors. The outputs are predicted surface quantities (pressure and wall shear stress) evaluated at a set of query points,

from which integral quantities such as drag can be computed using the same post-processing operators as for the CFD data.

Aerodynamic loads on athletes depend on both global posture (torso angle, head position, frontal area) and local interactions (arm/torso gaps, leg spacing, shoulder and helmet regions) that can alter separation and wake development. DoMINO captures this by combining a multiscale global encoding of the surface point cloud with local neighborhood features computed around discrete query points. For each query location, the model samples a local subregion, applies dynamic point-convolution kernels to extract geometry-aware features, and aggregates information from nearby points to predict the surface quantity of interest. The prediction is refined iteratively over multiple steps, which helps represent multiscale dependencies while maintaining a point-based formulation suitable for articulated postures.

The DoMINO configuration for this study used only the surface mode, as the primary objective was accurate prediction of drag via integrated surface pressure and shear stress. The configuration of the model has many hyperparameters, that could potentially be tweaked for optimal performance. The choice of hyperparameters aligns mainly with the default settings. A key setting is the latent interpolation resolution, which controls the spatial granularity of the learned global representation used when decoding predictions at query points. In the present configuration, the latent grid resolution was set to $128 \times 64 \times 48$, providing a relatively fine discretization of the domain in the streamwise, lateral, and vertical directions. This choice represents a practical trade-off. Higher latent resolutions can better preserve sharp spatial variations and geometry-induced gradients that influence surface pressure and wall shear stress, but increase memory footprint and training cost. Lower resolutions reduce cost but may over-smooth the latent representation. 200 000

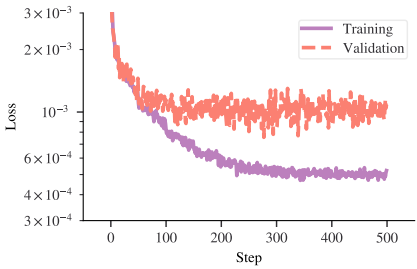


Fig. 2 Training and validation loss during optimization with all 240 simulations.

points were used to construct the dense global surface representation, while 8192 points were used for solution sampling. The local stencil construction used a radius of 0.05, with up to 64 neighbors. Surface neighborhood aggregation was enabled with 7 surface neighbors. The multiscale geometry encoder employed short- and long-range radii of 0.1 and 0.5, respectively, and the main multi-layer perceptron sub-networks used a width of 512. Training was run for 500 epochs, running on 4 NVIDIA A100 GPUs. The training itself took approximately 8 hours. Running the final trained model for a single geometry took less than a second.

3 Results

3.1 Training performance

Fig. 2 shows the training and validation loss during optimization. The training loss decreases steadily and stabilizes at approximately 5.0×10^{-4} in the final epochs. The validation loss follows a similar trend, reaching a level of around 1.0×10^{-3} in the final stages. The small gap between training and validation loss in the converged regime indicates good generalization, with no evidence of severe overfitting. Overall, the model shows stable convergence with well-aligned training and validation behavior.

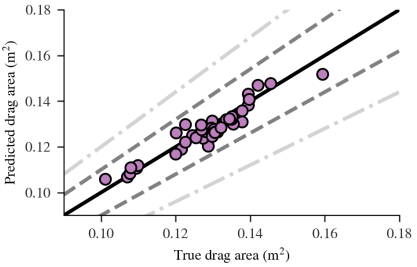


Fig. 3 Scatter plot of ground truth values vs predicted values in the test set. The dashed and dash-dotted lines represent 10 % and 20 % deviation, respectively.

3.2 Prediction performance

Fig. 3 compares the ground truth (CFD simulation) vs. the predicted drag area (ML surrogate model) in a scatter plot. The model reproduces the drag area with good agreement across all the test set geometries and positions, including the GCM geometry that was not part of the training dataset. All predictions fall within a ±10 % envelope, and the mean absolute percentage error is 2.69 %.

Fig. 4 compares the pressure coefficient distributions on the GCM geometry. While localized discrepancies are visible, most notably near the neck and upper-torso transition where small geometric or separation differences can produce sharp pressure gradients, the overall spatial patterns and magnitudes agree well.

3.3 Balance between geometries and positions

Since a significant bottle-neck in preparing the dataset is access to multiple athletes, it is of interest to compare the number of unique athlete geometries to the number of generated positions in terms of prediction accuracy of the trained surrogate model.

Tab. 1 shows the performance for 7 different combinations of number of unique geometries and number of generated positions per geometry. A clear trend emerges from this comparison: it is important to have as many unique

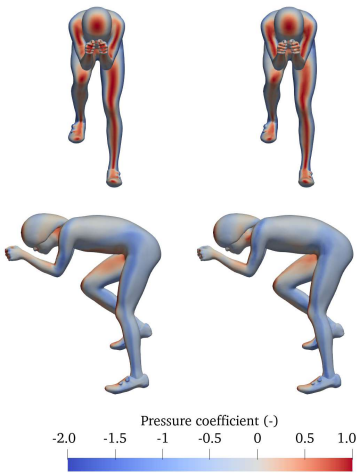


Fig. 4 Comparison of pressure coefficient between the ground truth CFD model (left) and the ML model (right) for the GCM model.

geometries as possible in the training dataset. In fact, having 12 unique geometries but only 5 positions (i.e. 60 geometries in total), gives better performance (3.94 %) than having 9 unique geometries and 20 positions (4.52 %). The data also suggest that the performance of the model could be further improved by increasing the number of unique geometries beyond the 12 geometries used in this dataset.

Table 1 Performance across designs for varying number of model, n_m and positions n_p. MAE: Mean absolute error. MAPE: Mean absolute percentage error.

Design	n_m	n_p	**MAE**	**MAPE (%)**
1	12	20	0.0035	2.69
2	9	20	0.0057	4.52
3	12	15	0.0042	3.32
4	6	20	0.0062	5.70
5	12	10	0.0046	3.66
6	3	20	0.0077	5.97
7	12	5	0.0051	3.94

4 Discussion

The results demonstrate that a modern point-cloud neural-operator surrogate developed for external aerodynamics can transfer effectively to flow over athletes, despite the markedly different nature of geometric variability. In contrast to common industrial benchmarks that rely on smooth morphing of a base surface, athlete posture variability is dominated by articulated limb motions.

A key practical contribution is the finding that the number of unique athlete geometries is more important than increasing the number of postures per athlete. This suggests that the dominant generalization challenge is not only the variation in poses for a fixed body, but the distribution shift introduced by different anthropometrics and local shape details. This conclusion is particularly relevant for sports engineering contexts where access to multiple elite athletes is limited.

Important limitations should be acknowledged. First, the CFD data are based on a certain computational setup, which constrains the surrogate to the uncertainties and biases of that modeling choice. The surrogate should therefore be interpreted as a fast emulator of this CFD configuration, rather than a general predictor of real-world performance. Second, the study only investigated prediction of surface quantities used for drag area prediction and did not attempt to reproduce full volume fields, which limits flow-physics interpretability. Third, equipment and geometric assumptions are constrained, for example the use of a single helmet model for all subjects, which may reduce robustness to other equipment choices. Finally, the dataset contains a limited number of unique scanned subjects, which is sufficient to demonstrate the viability of the approach, but not to claim broad population-level coverage. In particular, the model would be expected to only be applicable to time-trial positions and not more upright road positions.

These limitations motivate several directions for further work. Expanding the number of unique athlete geometries is the most direct path to improving accuracy and robustness. This could include other positions in cycling, or even other sports such as downhill skiing. Incorporating additional operating conditions, such as yaw angles, would improve practical relevance. A systematic comparison with alternative scalable surrogate paradigms would also be of interest.

Despite these limitations, the approach appears to be viable, especially for new application areas where fast results are more important than high accuracy. Combined with advances in pose estimation, these could, for instance, include drag area overlays on television broadcasts, bike fitting applications, and virtual wind tunnels.

5 Conclusion

A surrogate model trained on a CFD dataset generated with a virtual skeleton posture pipeline can predict cyclist drag area with good accuracy, including generalization to geometries not used during training. With 240 simulations in the dataset, the trained model reproduced drag area values within ± 10 % for all test cases and achieves a mean absolute percentage error of 2.69 %, while also capturing the dominant surface pressure patterns with only localized discrepancies in sensitive regions. A central practical finding is that increasing the number of unique athlete geometries improves surrogate accuracy more effectively than increasing the number of postures per athlete in the tested regimes. This work demonstrates that modern scalable external-aerodynamics surrogates can provide actionable, rapid posture evaluation for flow over athletes, opening the door for new workflows and application areas for sports aerodynamics.

Ethical approval. The athletes gave informed consent and the data collection was performed in accordance with the ethical guidelines of the host institution and the Declaration of Helsinki.

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